The DNA-Based Disease Diagnosis Model With Strand Displacement Reaction

Zhen Tang, *Member, IEEE*, Hongxv E, Chunlin Chen, Shiyin Li, Hongyan Feng, and Zhixiang Yin

***Abstract*—MicroRNAs (miRNAs) serve as crucial biomarkers in disease diagnosis. Although silicon-based electronic machine learning models provide efficient means for analyzing the massive data associated with miRNAs and for disease diagnosis, they remain constrained by low parallelism and poor biocompatibility. DNA computing provides an ideal interface for the integration of information technology and biotechnology. Here, we employ strand displacement reaction (SDR) to realize the DNA-based support vector machine (SVM) models for disease diagnosis that sequentially integrates four functional modules: weighted summation module, subtraction module, signal restoration module, and reporter module. In contrast to conventional silicon-based computing architectures, the DNA-based disease diagnosis model can directly and specifically identify the expression levels of target miRNAs from biological samples and drive model decisions in real time, enabling precise classification of disease states. We validate the disease diagnosis model using miRNAs expression levels, achieving diagnostic accuracies of 98.54% for cystic, mucinous and serous neoplasms (CMSN), 99.46% for Glioma, and 97.81% for clear cell carcinoma (CCC). The DNA-based disease diagnosis model simultaneously classifies the three disease states in parallel, and the results show a strong agreement with the actual disease states labeled in The Cancer Genome Atlas (TCGA). This work provides a new paradigm for engineering precise, intelligent and integrated disease diagnosis schemes at the molecular level.**

***Index Terms*—Disease diagnosis, DNA computing, strand displacement reaction, support vector machine.**

I. Introduction

MICRORNAS (miRNAs) are a class of single-stranded non-coding small RNA molecules encoded by endogenous genes in the cellular genome, which are involved in important biological activities such as cell proliferation, differentiation, apoptosis, and gene regulation [1]. MiRNAs and their expression levels are closely related to tumors and cardiovascular diseases, and they serve as significant biomarkers for the disease diagnosis [2]. However, individual miRNAs exhibit tremendous complexity in terms of both type and quantity. Currently, the clinical experimental methods for cancer diagnosis using miRNAs mainly include microarray analysis, quantitative polymerase chain reaction with reverse transcription (RT–qPCR), northern blot, and high throughput RNA sequencing [3]. However, these methods still have problems such as high cost, long time frames, and complex procedures. Moreover, the application of these techniques is highly dependent on professionals in the field, which restricts their use to specialized working environments.

The iterative updates of high throughput sequencing technologies and the completion of the Human Genome Project (HGP) have generated a vast amount of biomedical data, which provides rich data resources for the prediction of miRNA-disease associations (MDAs), miRNA-based disease diagnosis, and prognosis assessment [4]. Artificial intelligence, particularly machine learning, has become a powerful tool for disease diagnosis, supported by abundant data resources and characterized by its strong data processing and intelligent analysis capabilities. Many advanced models based on silicon-based electronic computers, such as deep learning, ensemble learning and graph neural networks, have made significant progress in using miRNA-related data for disease diagnosis and prognosis evaluation [5], [6], [7], [8], [9], [10], [11], [12], [13]. However, the performance of machine learning models under the silicon-based computing architecture is often constrained by low parallelism and poor biocompatibility. Specifically, there is a gap between silicon-based machine learning models and biological samples, the models cannot directly extract miRNAs expression levels from the biological samples and can only passively accept the miRNAs expression levels collected by humans as input. Manual acquisition of miRNAs expression levels from biological samples usually takes several hours or even days, especially when it involves the collection of multiple miRNAs expression levels, which increases the experimental and time costs. In addition, in the natural state, silicon-based machine learning models inherently lack parallelism and cannot perform multiple disease diagnosis tasks simultaneously, unless additional parallel computing architectures and hardware support are introduced. Therefore, it is important to implement machine learning models for disease diagnosis under a computing architecture with high parallelism and robust biocompatibility.

The development of molecular medicine and the rise of DNA computing have provided researchers with new research ideas. On the one hand, relevant work indicates that under the DNA computing architecture, various models and systems including Hopfield associative memory networks, machine learning, chaotic encryption systems, and convolutional neural networks can be engineered relatively maturely [14], [15], [16], [17],

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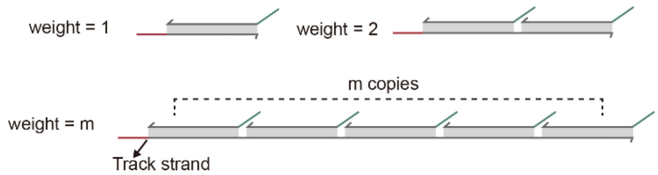

Fig. 1. The traditional design of weighted molecule.

[18], [19], [20]. On the other hand, DNA molecules can directly identify the miRNAs expression levels from biological samples and transmit it as an input signal into the model, making it an ideal interface for the integration of information technology and biotechnology [21], [22], [23]. In recent years, significant advancements have been made in the research of constructing support vector machine (SVM) models for disease diagnosis under the DNA computing architecture. For example, Seelig and colleagues introduced a molecular computing strategy for analyzing the information contained in complex gene expression characteristics without the need for expensive instruments and equipment. The workflow of this model began with training a computational classifier that uses labeled gene expression data, and this computational classifier is implemented at the molecular level to achieve expression analysis and classification of previously uncharacterized samples [24]. Sun and colleagues developed an alternative medical molecular computing approach called digital DNA strand displacement, which identifies targets and operates target valence through DNA polymerase-based extension and strand release. The model based on DNA polymerase significantly reduced the number of used oligonucleotide species, provided robust molecular classifiers [25]. However, these studies on DNA-based SVM models for disease diagnosis still have certain deficiencies. In most current enzyme-free models, the weight molecule in the weighted module is designed as a single long track strand and m copies of the same valence strand, where m denotes the weight value (Fig. 1). With the increase diversity of characteristic biomarkers and the expansion of model scale, this design is constrained by massive oligonucleotide consumption, limited synthesis and coding capacity, and thus cannot flexibly control the DNA implementation of weight values. Secondly, the subtraction module can lead to signal attenuation. In previous works, researchers often normalized experimental data to present the final output results. However, this step is time-consuming and increases the workload of the disease diagnosis process, hindering the realization of an intelligent integrated disease diagnosis solution. Furthermore, enzyme-mediated models introduce additional experimental complexity and biological enzyme costs.

Here, we employ strand displacement reaction (SDR) to realize the DNA-based SVM models for disease diagnosis that sequentially integrates four functional modules: weighted summation module, subtraction module, signal restoration module, and reporter module. The DNA-based disease diagnosis model can directly and specifically identify the expression levels of target miRNAs from biological samples and sequentially activates all functional modules, enabling precise classification of disease states. We initially validate the model performance by experimentally analyzing miRNAs expression data from patients with cystic, mucinous and serous neoplasms (CMSN) from The Cancer Genome Atlas (TCGA). To validate the parallelism of the model, we further perform experimental analysis using miRNAs expression data from patients with CMSN, Glioma, and clear cell carcinoma (CCC) from the TCGA. Visual DSD software is employed to perform all the simulation test experiments concerning the all DNA molecular reactions. The test results show that the classification results of the DNA-based disease diagnosis model show a strong agreement with the actual disease states labeled in TCGA. The designed model achieves the expected diagnostic performance, and shows robust feasibility and high parallelism in disease diagnosis, especially in the simultaneous diagnosis of multiple diseases. This work provides a new paradigm for engineering precise, intelligent and integrated disease diagnosis schemes at the molecular level.

The main contributions of this work are as follows:

1) The DNA-based disease diagnosis models with high parallelism and robust biocompatibility are established using enzyme-free SDR. Under the DNA computing architecture, the model can directly connect biological samples with disease diagnosis models, and simultaneously classify multiple disease states in parallel, achieving intelligent integrated diagnosis with accuracies of 98.54% for CMSN, 99.46% for Glioma, and 97.81% for CCC. This provides a superior comprehensive paradigm for precise disease diagnosis at the molecular level.
2) Utilizing a competitive cyclic mechanism, the weighted summation module achieves flexible control of weight values through adjusting the concentration ratio of the two DNA complexes. This approach avoids the complexity of designing valence strand copies and complex biological enzymes environment.
3) A signal restoration module is introduced to eliminate the subtraction module induced attenuation and deliver a uniform high intensity output signal. In DNA computing systems, the intrinsic characteristics of the subtraction module generate the signal attenuation and compromise output stability. For example, when the positive signal concentration is 100 nM and the negative signal concentration is 99 nM, only 1 nM survives subtraction, an attenuation of 99%. Such severe loss degrades detection sensitivity and limits computational accuracy. The signal restoration module compensates this loss by amplifying the remaining signal to a predefined high level, independent of the initial attenuation. It integrates seamlessly with existing DNA modules without extra redesign, the synergy of high gain, uniform output, and modular compatibility elevates the reliability and scalability of DNA-based system.

The remaining sections of this paper are as follows: materials and methods are given in Section II. Implementation of the DNA-Based SVM model in Section III. The DNA-based SVM model for disease diagnosis in Section IV. Conclusion is given in Section V.

II. Materials and Methods

A. Data Sources

The data used in this work is derived from TCGA and is available at https://www.cancer.gov/ccg/research/genome-sequencing/tcga. The work focuses on three common types of tumors: CMSN, Glioma and CCC. The CMSN dataset contains 807 tumor samples and 107 normal samples. The Glioma dataset contains 531 tumor samples and 95 normal samples. The CCC

TABLE I
THE MAIN PYTHON LIBRARIES

Function	Python libraries
Data preprocessing	Pandas numpy sklearn.impute sklearn.preprocessing
Feature extraction	scipy.stats sklearn.feature_selection.SelectFromModel
Modeling	sklearn.svm
Validation and evaluation	sklearn.model_selection sklearn.metrics
Visualization	Matplotlib pydotplus matplotlib.image

dataset contains 538 tumor samples and 73 normal samples. All samples are miRNA expression data. We partition each dataset into a training set and a validation set at a 7:3 ratio, ensuring that the model fully learns the data characteristics during training while enabling an objective assessment of its performance on unseen samples during validation.

B. Training the SVM Model in Silicon Electronic Computer

The core principle of SVM is to find a hyperplane (a straight line in two dimensions) to separate two types of data and maximize the shortest distance from this hyperplane to the two types of data (Margin). In the feature space, the hyperplane can be expressed as follows:

$$w^T x + b = 0 \tag{1}$$

where w is the normal vector of the hyperplane, and b is the bias. To maximize the margin, it is necessary to minimize the $\|w\|$ while meeting the constraint that all samples are correctly classified:

$$\min_{w,b} \frac{1}{2}\|w\|^2$$
$$s.t.\ \ y_i(w^T x_i + b) \geq 1,\ \ \forall i \tag{2}$$

Notably, the weighted summation module has the capability to sense miRNA expression levels in biological samples and assigns weights accordingly, whereas the bias does not participate in this weighted summation process. Therefore, during the DNA-based implementation of the SVM, the bias value is mapped to the concentration of the weighted summation output signal molecules D or H, which together with the weighted summation output participated in subsequent subtraction module, signal restoration module, and reporter module reactions. Specifically, when the bias b>0, its value is mapped to the concentration of the positive weighted summation output signal D; when the bias b<0, its value is mapped to the concentration of the negative weighted summation output signal H.

We completed the entire process of data preprocessing and model training based on the Python (version 3.8), covering several aspects such as data preprocessing, feature extraction, model construction, performance evaluation, and result visualization. The main Python libraries used are shown in Table I.

C. Evaluation Metrics

To evaluate the performance of the SVM models in disease classification tasks, six evaluation metrics are employed: Accuracy, Sensitivity, Recall, Specificity, Precision and F1-score. The meanings and formulas of these evaluation metrics are as follows:

Accuracy: it measures the proportion of the model's overall predictions being correct, including two situations: predicting positive cases as positive and identifying negative cases as negative. The mathematical formula is:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

Sensitivity/Recall: they measure the model's ability to identify true positive samples (diseased samples). The mathematical formula is:

$$\text{Sensitivity/Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

Specificity: it measures the model's ability to identify true negative samples (healthy samples). The mathematical formula is:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{5}$$

Precision: it measures the proportion of the model's positive samples predictions being correct. The mathematical formula is:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

F1 score: it is the harmonic mean of Precision and Recall. The mathematical formula is:

$$\text{F1} - \text{score} = 2 \times \frac{\text{Pr ecision} \times \text{Recall}}{\text{Pr ecision} + \text{Recall}} \tag{7}$$

Among these formulas, TP, TN, FP, and FN represent true positive (TP), true negative (TN), false positive (FP), and false negative (FN) respectively. Among the above six evaluation metrics, Sensitivity and Specificity serve as the core indicators in disease diagnosis, reflecting the minimization of missed and misdiagnosed cases, respectively; Precision and F1-score further quantify the reliability of positive predictions; Accuracy is an overall performance indicator. The integrated use of these metrics enables multivariate assessment of the suitability and robustness of the SVM model for disease diagnosis.

D. SDR

SDR represents one of the most frequently employed biotechnologies for DNA computing systems, driven by its enzyme-free, programmable, predictable products, and simple experimental conditions [26], [27], [28], [29], [30], [31]. SDR is a DNA hybridization reaction mediated by toehold. The input strand is a long DNA single strand that is completely complementary to the DNA substrate. It specifically recognizes and attaches to the toehold suspended by the partially complementary DNA substrate, and releases the originally bound strand on the DNA substrate in entropy-driven manner. Fig. 2(a) shows the principle of SDR [32], [33]. Here, the DNA substrate has only one toehold. After the SDR occurs, the waste species with a stable double helix structure is produced, and its toehold is in a non-exposed state. The entire reaction process is irreversible. Irreversible

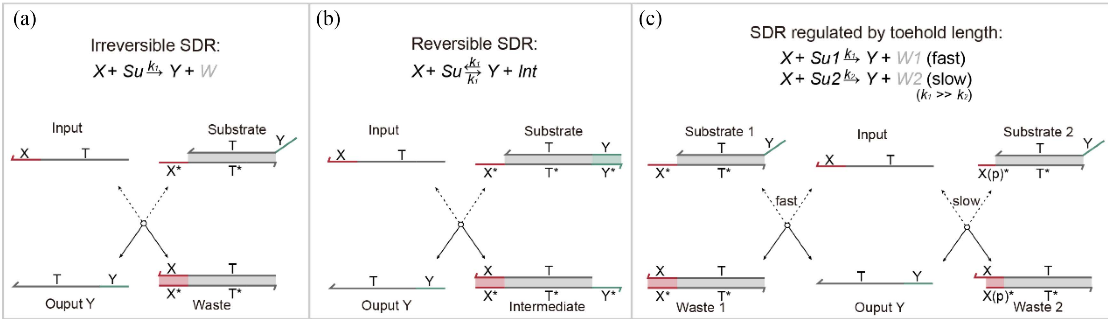

Fig. 2. The principle of SDR. (a) Irreversible SDR. (b) Reversible SDR. (c) SDR regulated by toehold length.

SDR can be described by the following DNA chemical reaction network:

$$X + Su \xrightarrow{k_1} Y + W \tag{8}$$

where k_1 represents the reaction rate.

Reversible SDR can be achieved through the toehold exchange mechanism, as shown in Fig. 2(b). A classic application case of reversible SDR based on toehold exchange mechanism is the DNA seesaw circuit system proposed by Qian et al. in 2011 [34]. Reversible SDR can be described by the following DNA chemical reaction network:

$$X + Su \underset{k_1}{\overset{k_1}{\longleftrightarrow}} Y + Int \tag{9}$$

The rate of SDR can be regulated by adjusting the length of the toehold, as shown in Fig. 2(c) [35]. The toehold X(p)$*$ of DNA substrate 2 is a truncated version of the toehold X$*$ of DNA substrate 1, which makes the rate at which the input DNA strand binds to substrate 1 much greater than the rate at which it binds to substrate 2. It can be described by the following DNA chemical reaction network:

$$\begin{aligned} X + Su1 &\xrightarrow{k_1} Y + W1 \\ X + Su2 &\xrightarrow{k_2} Y + W2 \end{aligned} \tag{10}$$

where k_1 and k_2 represent the reaction rate, $k_1 >> k_2$.

E. Visual DSD

Visual DSD is a powerful tool for analyzing and visualizing DNA computing systems based on SDR, and is widely used in the design, simulation and analysis of DNA computing systems. The software was initially proposed by Andrew Phillips et al., and then its modeling and biochemical experimental data were fully expanded [36], [37], [38]. In this work, we used the Visual DSD software to conduct all the simulation test experiments of DNA molecular reactions. The software version used was “v2015-0325”. To comprehensively analyze the SDR in this work and evaluate the performance of each functional module and model, the main parameters of the Visual DSD programs are set in Table II, which are able to simulate all possible reactions between DNA molecules.

TABLE II
THE MAIN PARAMETERS OF THE VISUAL DSD PROGRAMS

Species	Parameters
The lengths of toeholds	6 nt
The length of the truncated version toeholds	3 nt
The lengths of recognition domains	20 nt
Binding and unbinding rates	0.03 s^{-1} for the binding rate 0.1 s^{-1} for the unbinding rate
Directive	simulator deterministic compilation infinite

III. Implementation of the DNA-Based SVM Model

In this section, we describe the details of implementing the DNA-based SVM model using SDR, which sequentially integrates four functional modules: weighted summation module, subtraction module, signal restoration module, and reporter module. In 2000, Yurke and colleagues employed the toehold-mediated SDR to construct a DNA nanomachine shaped like tweezers, and interest in SDR subsequently surged [32]. In 2011, Qian and colleagues first implemented a Hopfield associative memory using the weighted and summation module based on SDR [39]. In 2018, Qian further systematized the weighted mechanism and, by cascading weight multiplication, summation, pairwise annihilation, signal restoration and reporting modules, experimentally demonstrated the winner-take-all neural network [14]. Such DNA-based weighted mechanisms commonly adopt binary 0/1 weighted, including DNA-based convolutional neural network [15], [40]. However, in DNA-based classification models for disease diagnosis, the weighted module specifically senses the expression levels of target miRNAs and assigns numerical weights to these molecules to capture their features (such as Logistic regression model and SVM). The weighted module evolves from binary 0/1 weighted to numerical weighted and most studies replicate the valence strand to encode the weight [22], [41]. In this section, we use a competitive cycle mechanism to realize numerical weighted. Next, we provide a systematic explanation of the design of the four functional modules and employ Visual DSD software to evaluate the feasibility and stability of each individual modules and of the integrated cascade.

A. Weighted Summation Module

In the design of the weighted summation module, we use a competitive cycle mechanism based on SDR to implement the weighted summation operation, where the weight values are mapped to the concentration ratio of the two DNA complexes. The weighted summation module consists of the amplification molecule (Amp $\mathrm{W_i}$ or $\mathrm{W_j}$), the inhibition molecule (Inh $\mathrm{I_i}$ or $\mathrm{I_j}$) and the fuel molecule FW or FV, where the subscript i or j represents the numbers of different variables (Fig. 3(a)). In the weighted summation module, each variable corresponds to a unique amplification molecule (Amp $\mathrm{W_i}$ or $\mathrm{W_j}$), and the inhibition molecule (Inh $\mathrm{I_i}$ or $\mathrm{I_j}$). The fuel molecules participate in the reaction after the molecules representing the variables specifically bind to the amplification molecules. To minimize the number of DNA molecule types as much as possible, only two types of fuel molecules are needed. Among them, fuel molecule FW is used for positive weighting, and fuel molecule FV is used for negative weighting. The weighted summation module can be described by the following DNA chemical reaction network:

$$X_i + W_i + I_i + FW \xrightarrow{k_1} D + W \ (X_i \text{ for positive weight}) \tag{11}$$

or

$$X_j + W_j + I_j + FV \xrightarrow{k_1} H + W \ (X_j \text{ for negative weight}) \tag{12}$$

When the single-stranded DNA $\mathrm{X_i}$ representing the variable is input into the weighted summation module, the single-stranded $\mathrm{X_i}$ binds to the Amp $\mathrm{W_i}$ and the Inh $\mathrm{I_i}$ with equal probability because both molecules have the same toehold $\mathrm{X_i}*$. The single-stranded $\mathrm{X_i}$ undergoes an irreversible SDR with the Inh $\mathrm{I_i}$, and the reaction terminates after the generation of waste species. Meanwhile, the single-stranded $\mathrm{X_i}$ reacts with the Amp $\mathrm{W_i}$ to generate a weighted output (the positive weighted output signal is represented by the letter D, and the negative weighted output signal is represented by the letter H) and an intermediate species. The intermediate binds to fuel FW via toehold $W*$, producing waste species and recycling the single-stranded $\mathrm{X_i}$. Initial concentrations of single-stranded $\mathrm{X_i}$, Amp $\mathrm{W_i}$, Inh $\mathrm{I_i}$ and fuel molecule FW are respectively symbolized by $[\mathrm{X_i}]_0$, $[\mathrm{W_i}]_0$, $[\mathrm{I_i}]_0$ and $[\mathrm{FW}]_0$, with $[\mathrm{FW}]_0 >> [\mathrm{W_i}]_0$, $[\mathrm{I_i}]_0 >> [\mathrm{X_i}]_0$. After the first round of the cycle, the concentration of the weighted output signal is:

$$[\mathrm{Output}]_1 = [\mathrm{X}_i]_0 \times \frac{[\mathrm{W}_i]_0}{[\mathrm{W}_i]_0 + [\mathrm{I}_i]_0} \tag{13}$$

The concentration of the recycled single-stranded $\mathrm{X_i}$ is:

$$[\mathrm{X}_i]_1 = [\mathrm{X}_i]_0 \times \frac{[\mathrm{W}_i]_0}{[\mathrm{W}_i]_0 + [\mathrm{I}_i]_0} \tag{14}$$

After t rounds of cycle, the concentration of the weighted output signal is:

$$\begin{aligned} \lim_{t\to\infty} [\mathrm{Output}]_\mathrm{t} &= \lim_{t\to\infty} \sum_{t=1}^{t} [\mathrm{X}_i]_0 \times \left(\frac{[\mathrm{W}_i]_0}{[\mathrm{W}_i]_0 + [\mathrm{I}_i]_0} \right)^t \\ &= [\mathrm{X}_i]_0 \times \frac{[\mathrm{W}_i]_0}{[\mathrm{I}_i]_0} \end{aligned} \tag{15}$$

Overall, the single-stranded $\mathrm{X_i}$ with an initial concentration of $[\mathrm{X_i}]_0$ can generate the weighted output signal with a concentration of $[\mathrm{X_i}]_0 \times [\mathrm{W_i}]_0/[\mathrm{I_i}]_0$ through the weighted summation module, and the weight values are mapped to the concentration ratio of the Amp $\mathrm{W_i}$ to the Inh $\mathrm{I_i}$. This design only requires adjusting the concentration ratio of Amp $\mathrm{W_i}$ and Inh $\mathrm{I_i}$ to flexibly set any weight value. The weight value, weighted of a variable, and weighted summation of multiple variables of the weighted summation module can be expressed by the following formulas:

$$\begin{aligned} &\mathrm{weight} : w_i = \frac{[\mathrm{W}_i]_0}{[\mathrm{I}_i]_0} \\ &w_1 \times x_1 = [\mathrm{X}_1]_0 \times \frac{[\mathrm{W}_1]_0}{[\mathrm{I}_1]_0} \\ &\sum_{i=1}^{n} w_i \times x_i = \sum_{i=1}^{n} [\mathrm{X}_i]_0 \times \frac{[\mathrm{W}_i]_0}{[\mathrm{I}_i]_0} \end{aligned} \tag{16}$$

Next, we first verify the weighted of a variable, that is, $y = w_1 \times x_1$. We employ Visual DSD software to test the weighted summation module with four distinct weight sets, and all the outputs moved to the correct weighted signals (Fig. 4(a)). Then, we verify the weighted summation of three variables. We assign two different weight combinations to the three variables. In the first combination, we set $(x_1, x_2, x_3) = (1.5, 2, 2)$ and adopt $(w_{11}, w_{21}, w_{31}) = (2, 2, 2.5)$; the weighted summation output is $y = w_{11} \times x_1 + w_{21} \times x_2 + w_{31} \times x_3$. In the second combination, we set $(x_1, x_2, x_3) = (1, 1, 2)$ and adopt $(w_{12}, w_{22}, w_{32}) = (2, 2, 3)$; the weighted summation output is $y = w_{12} \times x_1 + w_{22} \times x_2 + w_{32} \times x_3$. We employ Visual DSD software to test the weighted summation module with two distinct weight sets, and all the outputs moved to the correct weighted summation signals (Fig. 4(b)). Fig. 4(c) shows the entire SDR network of the weighted summation module extracted from Visual DSD. In the SDR network, besides the expected products (species Intermediate, Waste, and Output), some unwanted products (species sp6, sp7 and sp9) are also generated. These unwanted products are transient states formed by brief binding, and as the reaction proceeds, the overall pathway strictly follows the designed reaction pathways. These test results show that the weighted summation module has been successfully implemented and has good feasibility.

B. Subtraction Module, Signal Restoration Module and Reporter Module

The function of the subtraction module is to annihilate the positive weighted signal D and the negative weighted signal H produced by the weighted summation module in pairs. When the concentration of the positive weighted signal D exceeds that of the negative weighted signal H, the subtraction module leaves the excess of positive weighted signal D that proceeds to downstream reactions; otherwise, residual the negative weighted signal H continues. We designed a subtraction molecule (Sub) to achieve the function of the subtraction module, which is a DNA complex formed by the hybridization of a long DNA single strand and two short DNA single strands with complementary bases (Fig. 3(b)). The subtraction module can be described by the following DNA chemical reaction network:

$$D + H + Sub \xrightarrow{k_1} W \tag{17}$$

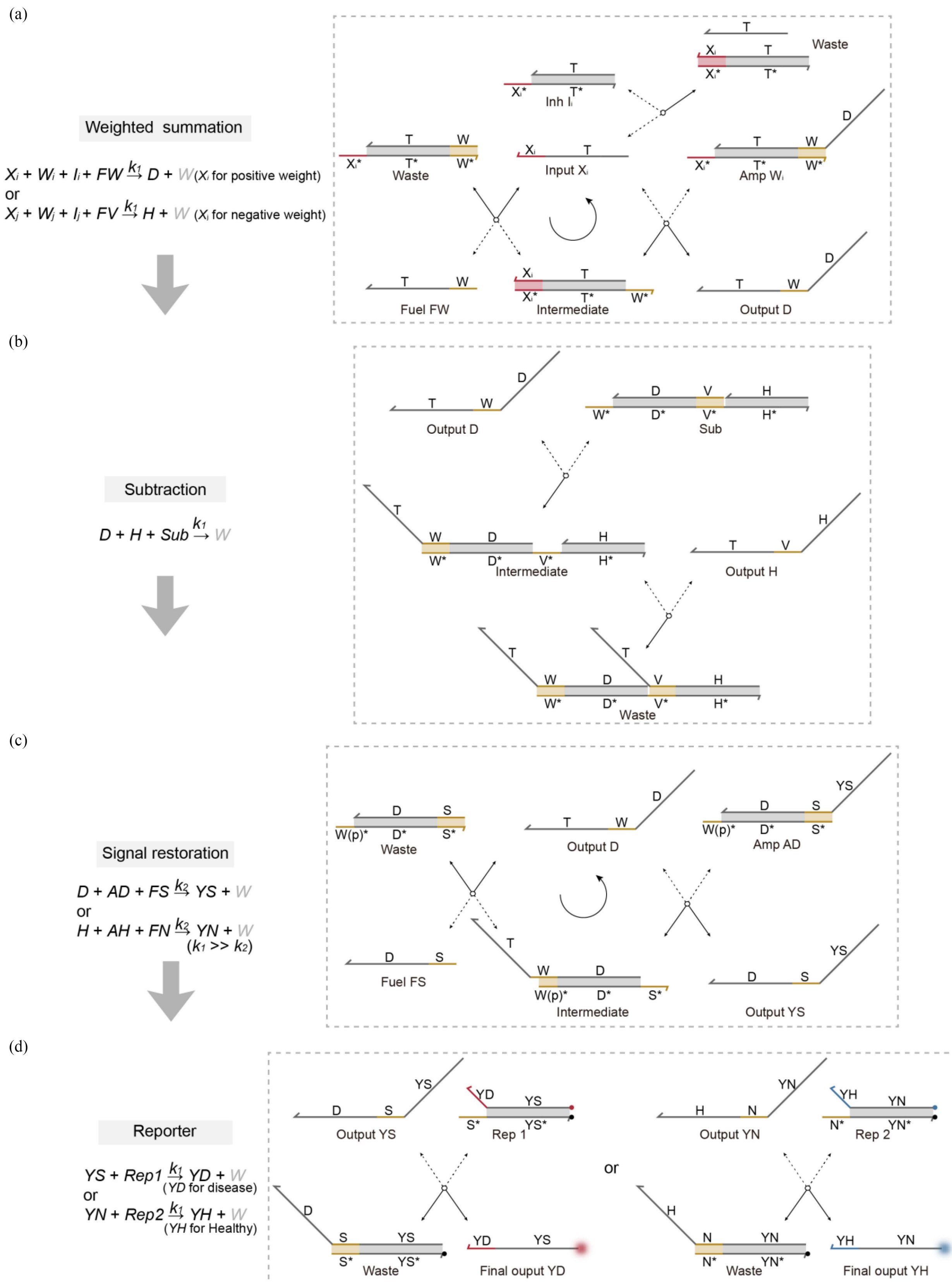

Fig. 3. Four functional modules. (a) Weighted summation module. (b) Subtraction module. (c) Signal restoration module. (d) Reporter module.

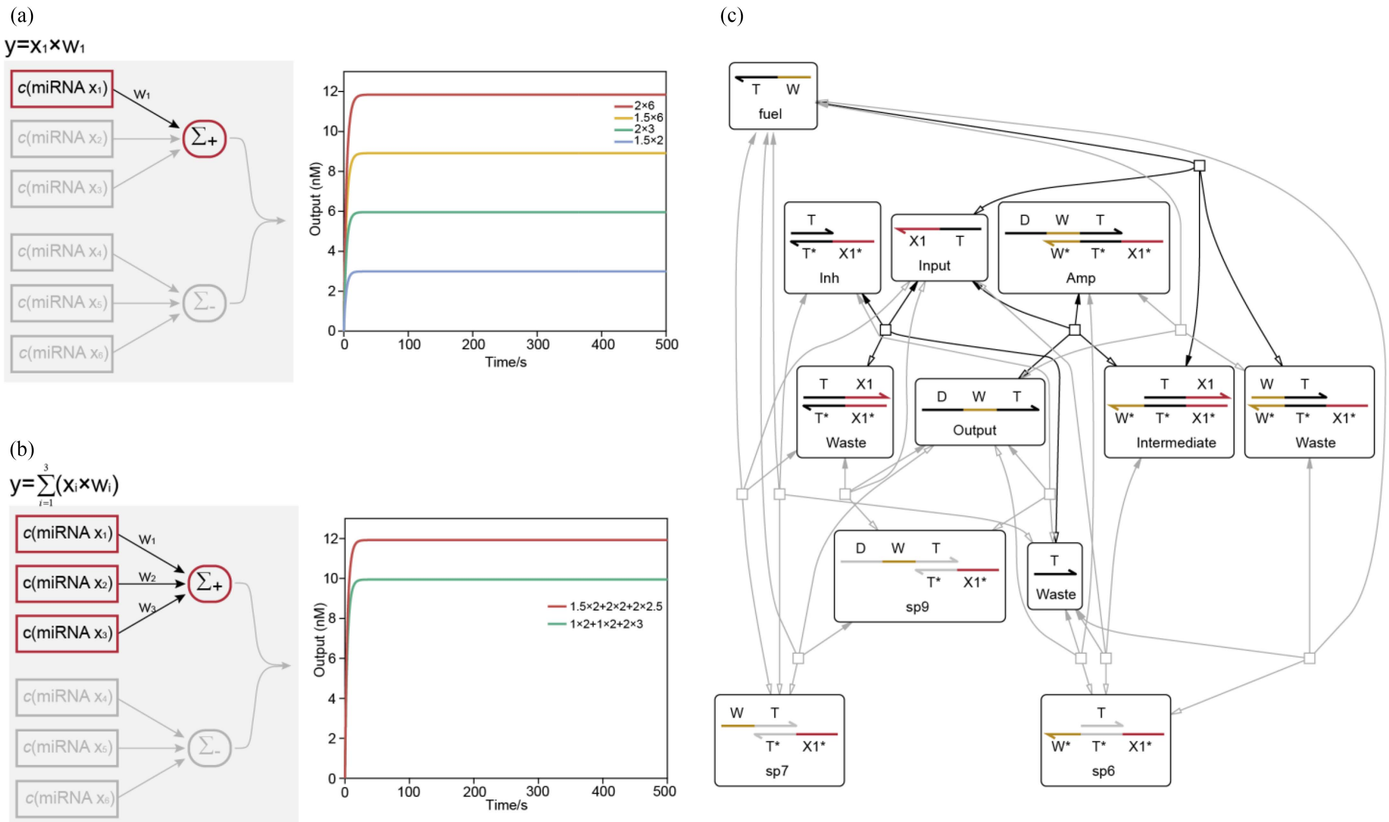

Fig. 4. Testing of the weighted summation module. (a) Testing of the weighted of a variable. (b) Testing of weighted summation of three variables. (c) Entire SDR network of the weighted summation module.

The positive weighted signal D first binds specifically to the toehold W$*$ of the molecule Sub, producing an intermediate species and exposing the toehold V$*$. The negative weighted signal H in turn binds specifically to the exposed toehold V$*$ of the molecule Sub to generate the waste species, and the signals D and H annihilate in pairs. After pairwise annihilation, the remaining weighted signal reacts with the reporter module to produce the final output signal. We use Visual DSD to test the subtraction module with two different combinations of positive and negative weighted signals. The test results show that the subtraction module can perform pairwise annihilation well (Fig. 5(a)). In the absence of the signal restoration module, the remaining weighted signal after going through the subtraction module directly generates the final output signal through the reporter module. It can be clearly observed that although the subtraction module has achieved the expected function, different degrees of signal attenuation occur due to the different concentrations of positive and negative weighted signals. This signal attenuation affects the stability of the output signal and interferes with the detection of the output signal and the accuracy of the model.

Therefore, we further designed a signal restoration module to address the signal attenuation issue caused by the subtraction module, which can restore different degrees of signals to the uniform high intensity output signals. The signal restoration module consists of an amplification molecule (Amp AD or AH) and the molecule strand (FS or FN) (Fig. 3(c)). The signal restoration module can be described by the following DNA chemical reaction network:

$$D + AD + FS \xrightarrow{k2} YS + W \tag{18}$$

or

$$H + AH + FN \xrightarrow{k2} YN + W \tag{19}$$

where the reaction rate $k_1 >> k_2$.

Take the positive weighted signal D as an example. If the remaining signal after the subtraction module is the positive weighted signal D, it first reacts with the amplification molecule AD to produce the output signal YS and the intermediate species, the intermediate is then further combined with the fuel FS via the toehold S$*$ to produce the waste species and recycling the signal D. The cyclic reaction continues until the molecule AD is depleted. In the signal restoration module, the initial concentration of the fuel FS is much greater than that of the molecule AD, that is, $[FS]_0 >> [AD]_0$. It is worth noting that an important factor to be considered when designing the signal restoration module is how to ensure that the weighted signal goes through the subtraction module before the signal restoration module. Our solution is that the toehold length of the amplification molecule of the signal restoration module is half that of the toehold of the subtraction molecule, which makes the bind rate k_1 of the weighted signal to the subtraction molecule much greater than the bind rate k_2 of the weighted signal to the amplification molecule of the signal restoration module, ensuring that the weighted signal must be preferentially

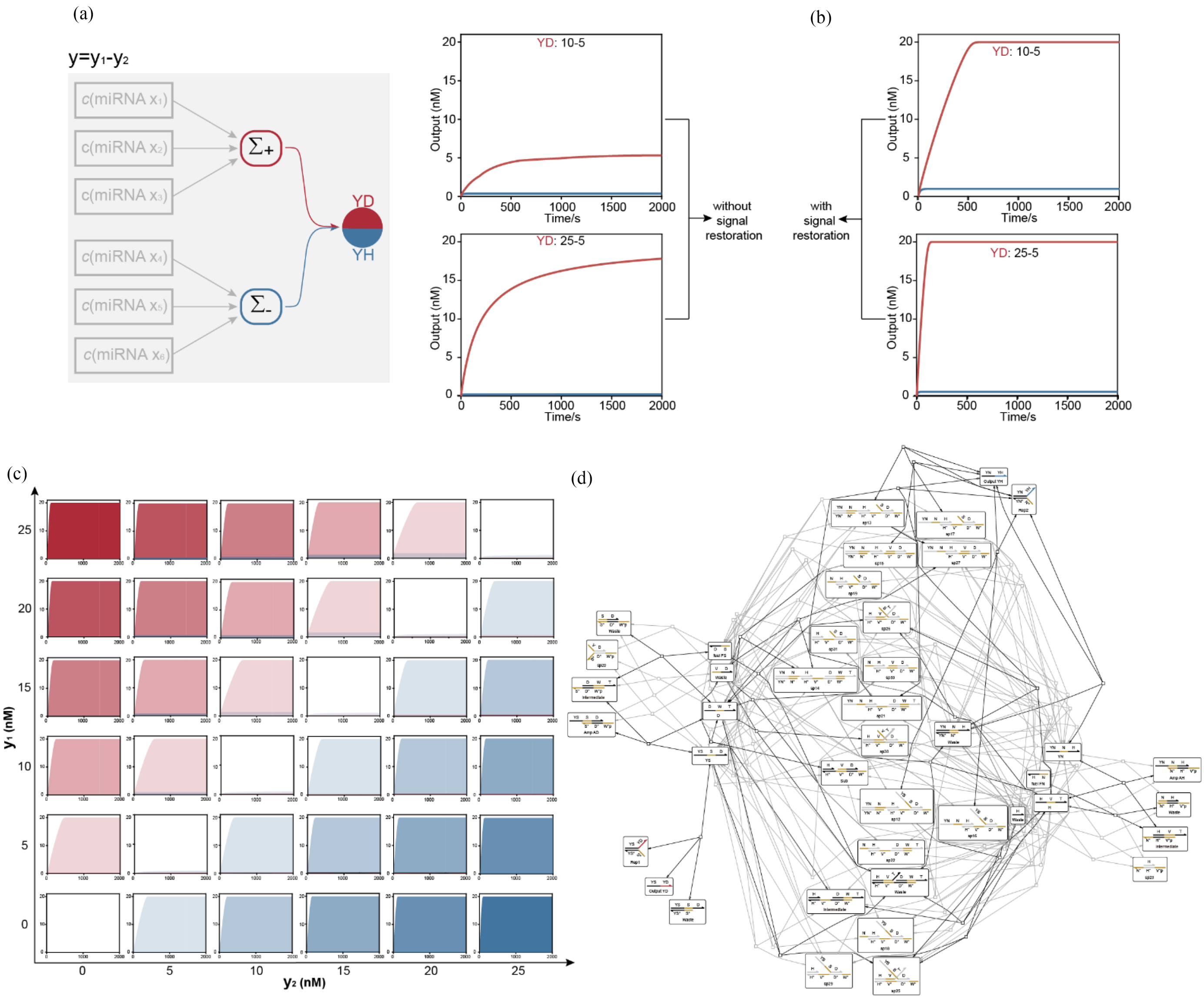

Fig. 5. Testing of the subtraction module, signal restoration module and reporter module. (a) The subtraction module and the reporter module are tested without the signal restoration module. (b) The subtraction module and the reporter module are tested with the signal restoration module. (c) Testing of 36 different combinations with the signal restoration module. (d) Entire SDR network of the subtraction module, signal restoration module and reporter module.

combined with the subtraction molecule, and then go through the signal restoration module. Thus, the low concentration weighted signal D can produce the high concentration output signal YS through the signal restoration module. The high concentration output signal YS is further used to generate the final output signal through the reporter module (Fig. 3(d)). After using the signal restoration module, the test results using Visual DSD show that the final output signals can maintain high strength (Fig. 5(b)). The reporter module can be described by the following DNA chemical reaction network:

$$YS + Rep1 \xrightarrow{k1} YD + W \tag{20}$$

or

$$YN + Rep2 \xrightarrow{k1} YH + W \tag{21}$$

To further evaluate the performance of the model after introducing the signal restoration module, we test 36 different combinations of positive weighted signals (ranging from 0 nM to 25 nM) and negative weighted signals (ranging from 0 nM to 25 nM) using Visual DSD. When the concentrations of the positive weighted signal and the negative weighted signal are close, the signals annihilate in pairs in the subtraction module, with almost no remaining signals transmitted to the signal restoration module and the reporter module. As a result, no final output signal is generated, as shown diagonally in Fig. 5(c). When the concentrations of the positive weighted signal and the negative weighted signal are different, the signals annihilate in pairs in the subtraction module, and the remaining signals are transmitted to the signal restoration module. Although the remaining signals vary due to the difference in positive and negative weighted signal concentrations, they can all undergo the signal restoration to produce the high concentration output

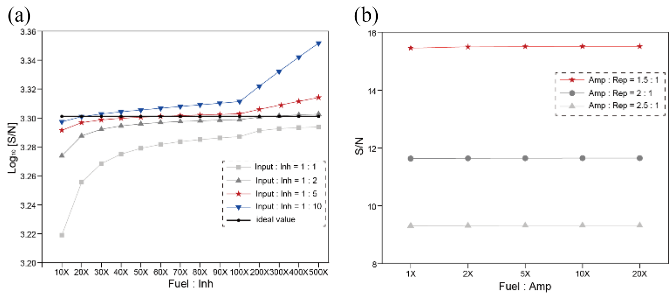

Fig. 6. SNR analysis. (a) SNR analysis of the weighted summation module. (b) SNR analysis of the entire reaction network.

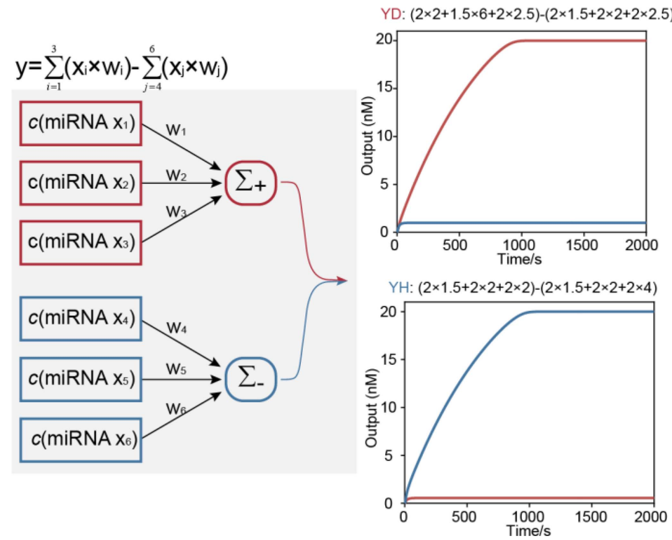

Fig. 7. Modules cascade testing.

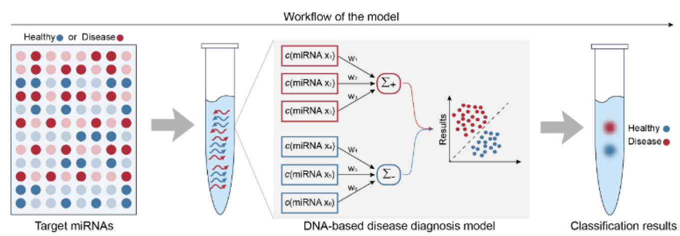

Fig. 8. The workflow of the DNA-based SVM model.

signal, which is read out by the reporter module. As shown on both sides of the diagonal in Fig. 5(c), different degrees of concentration difference can be output in a uniform high intensity signal. Fig. 5(d) shows the entire SDR network of the subtraction module, signal restoration module and reporter module extracted from Visual DSD. These test results show that the signal restoration module can effectively overcome the signal attenuation, and the designed subtraction module, signal restoration module and reporter module have good feasibility and cascade.

C. Modules Cascade Testing

After completing the above feasibility verification, we systematically optimize the entire reaction network via signal-to-noise ratio (SNR) analysis. Figs. 4(c) and 5(d) demonstrate that unwanted products and reaction pathways exist in the entire SDR network, mainly arising from slight reaction leakage and transient intermediates. Rational regulation of the concentration ratios among functional modules effectively suppresses noise interference. We first perform SNR analysis for the weighted summation module, which consists of input, inhibition, amplification, and fuel molecules. Since inhibition and amplification molecules are set at a weighted ratio, we focus on the concentration relationships among input, inhibition, and fuel molecules. To ensure sufficient module response to input molecules, input-to-inhibition concentration ratios of 1:1, 1:2, 1:5, and 1:10 are examined. Fuel molecules drive cyclic reactions and require higher concentrations than other components; thus, 14 fuel concentration gradients are tested under each of the four input-inhibition ratios. SNR analysis reveals that an input-to- inhibition -to-fuel ratio of 1:10:200 achieves optimal noise suppression and maximum SNR (Fig. 6(a)). Following determination of the optimal concentration ratio for the weighted summation module, we optimize the SNR of the entire reaction network. In the subtraction module, subtraction molecules are used in excess to ensure sufficient pairwise annihilation of positive and negative signals. We therefore focus on optimizing concentration ratios of amplification and fuel molecules in the signal restoration module, and reporter molecules in the reporter module. To yield a uniform high intensity output signal, the concentration of the reporter molecule is set at 20 nM, with amplification-to-reporter ratios of 1:1, 1:2, 1:5, and 1:10. Five fuel- amplification concentration combinations are tested for each ratio. The result shows that a reporter-to-amplification-to-fuel ratio of 1:1.5:15 effectively suppresses noise interference and enables favorable SNR performance across the entire reaction network (Fig. 6(b)).

We further test the cascading of these functional modules. Taking a 1×6 input pattern encoded by six DNA input strands (X_1, X_2, X_3, X_4, X_5, X_6) as an example, that is, $y = w_1 \times x_1 + w_2 \times x_2 + w_3 \times x_3 – w_4 \times x_4 – w_5 \times x_5 \text{-} w_6 \times x_6$. In the 1×6 input pattern, the x_1, x_2, and x_3 variables are set to positive weights, and the x_4, x_5, and x_6 variables are set to negative weights. We use Visual DSD to test the performance of the module cascade with two different combinations of variables and weights as input (Fig. 7). As expected, regardless of whether the positive weighted signal is larger than the negative weighted signal or the negative weighted signal is larger than the positive weighted signal, the DNA-based SVM model formed by the cascading of the four functional modules can correctly respond within a relatively short time span and output the uniform high intensity signal. The test results show that this is a model with low leakage, high signal fidelity, robust feasibility and strong stability.

IV. The DNA-Based SVM Model for Disease Diagnosis

In this section, we use the designed DNA-based SVM model for disease diagnosis. The DNA-based SVM model directly and specifically identifies the expression levels of target miRNAs from biological samples, benefiting from the good biocompatibility of the model, and sequentially activates all functional modules, enabling precise classification of disease states. The entire workflow of the model is shown in Fig. 8. We initially use the model to classify the disease states of CMSN to preliminarily

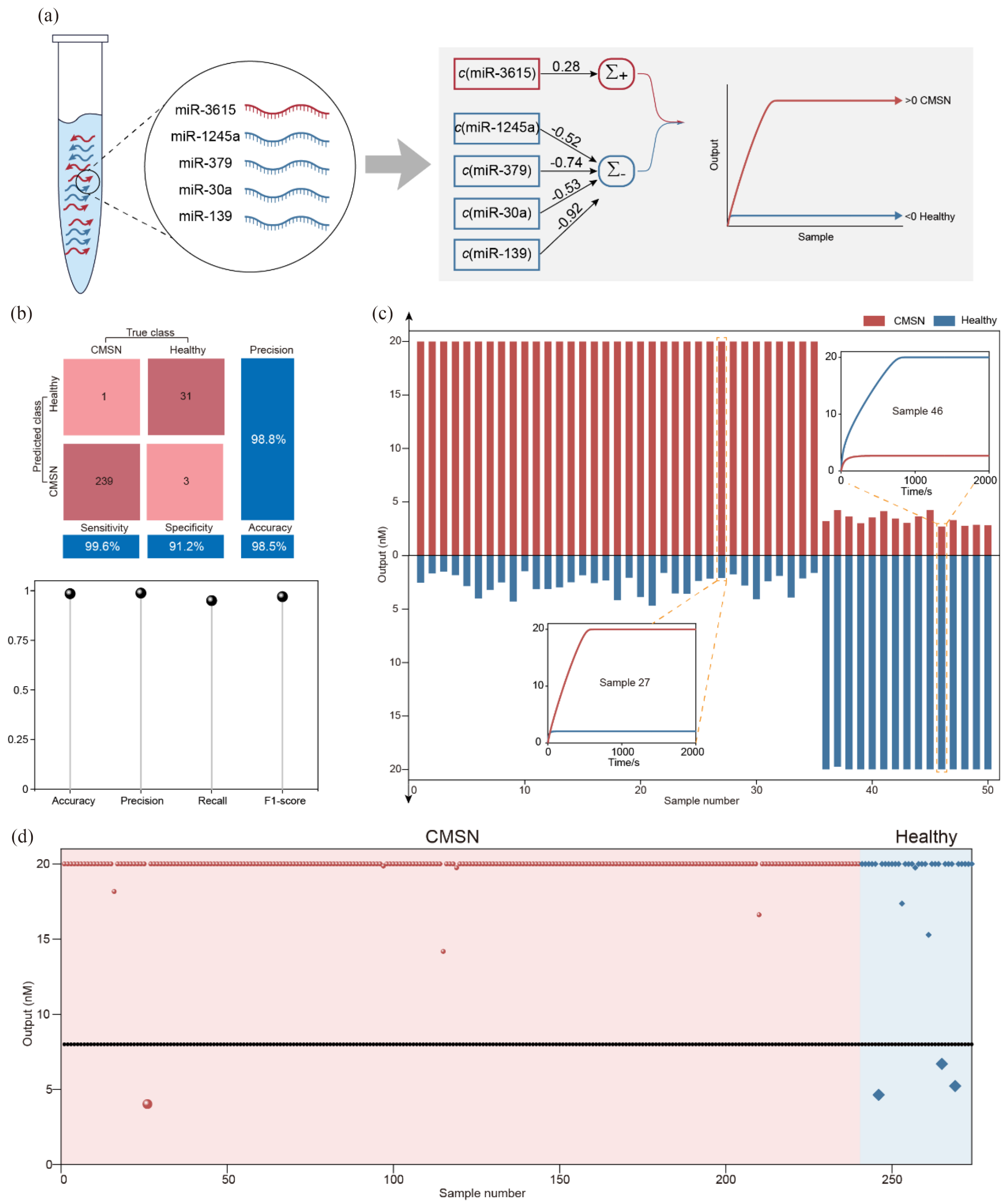

Fig. 9. The DNA-based SVM model for CMSN. (a) Training the SVM model in silicon electronic computer. (b) Evaluation metrics of the model. (c) Disease states classification of 50 random samples using the DNA-based SVM model. (d) CMSN disease State classification of 274 samples (test set) using the DNA-based SVM model.

verify the feasibility of the model in disease diagnosis. To validate the parallelism of the model, we further perform the simultaneous diagnosis of CMSN, Glioma, and CCC. Visual DSD is used to perform all the relevant experimental tests.

A. The DNA-based SVM Model for CMSN

To validate that the designed DNA-based SVM model can be effectively applied to disease diagnosis, we apply it to classify the disease states of CMSN. We first trained the SVM model on a silicon electronic computer using the miRNAs expression level data of CMSN retrieved from TCGA (807 tumor samples and 107 normal samples), and extracted the disease-relevant characteristic miRNAs (miRNA-3615, miRNA-1245a, miRNA-379, miRNA-30a and miRNA-139) together with their associated weight values (0.28, -0.52, -0.74, -0.53 and -0.92), as shown in Fig. 9(a). In the testing phase, the model correctly classified 239 CMSN samples and 31 healthy samples and only 1 CMSN sample was misclassified as healthy, and three healthy samples were misclassified as CMSN (Fig. 9(b)). We calculated six main evaluation metrics of the model: Sensitivity was 99.6%, Specificity was 91.2%, Precision was 98.8%, Accuracy was 98.5%,

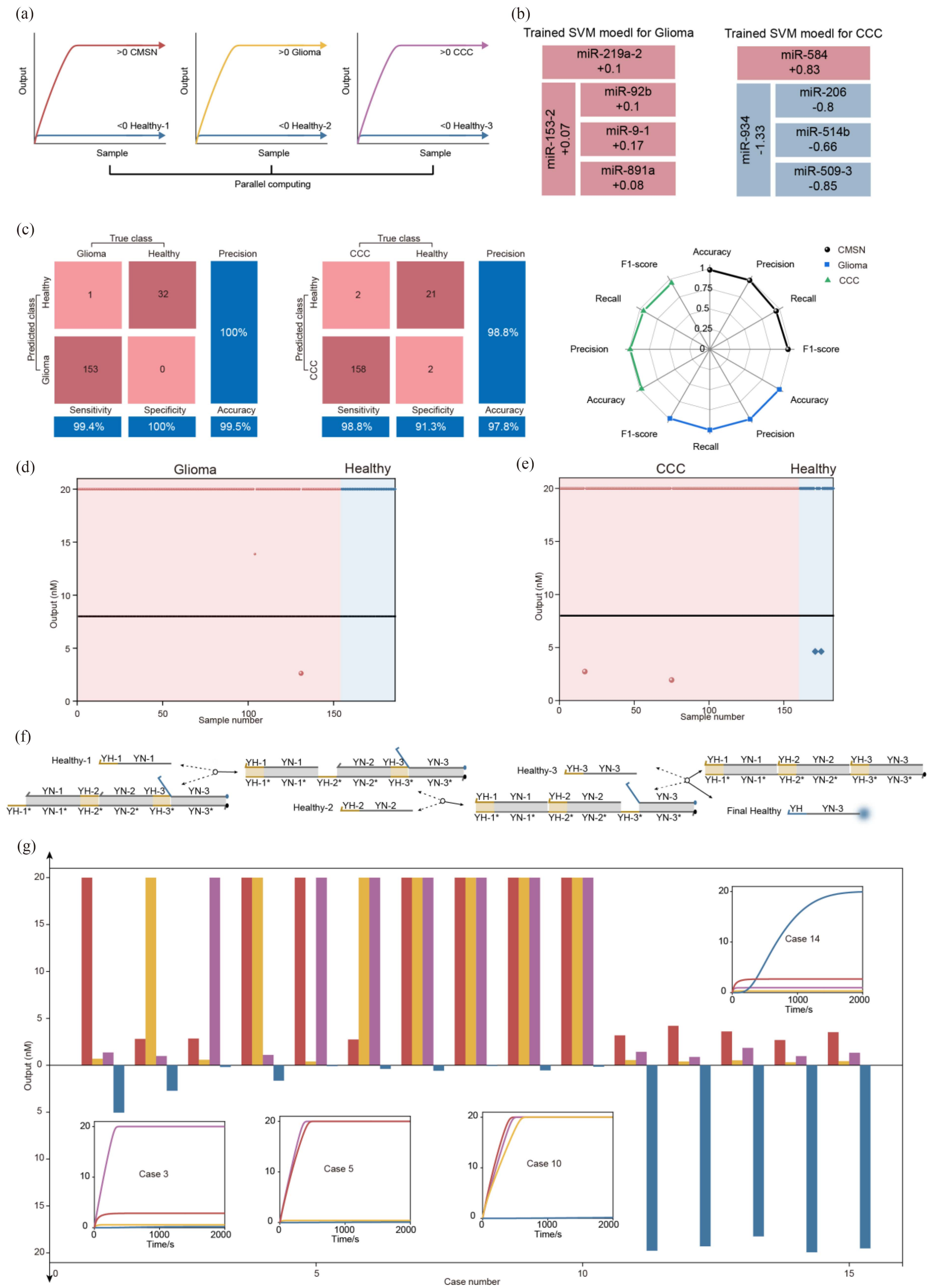

Fig. 10. Parallel disease diagnosis using the DNA-based SVM models. (a) Parallel diagnosis of CMSN, Glioma, and CCC. (b) Training two SVM models in silicon electronic computer for Glioma and CCC, respectively. (c) Evaluation metrics of the models. (d) Glioma disease State classification of 186 samples (test set) using the DNA-based SVM model. (e) CCC disease State classification of 183 samples (test set) using the DNA-based SVM model. (f) Three-input DNA AND gate. (g) The parallel classification results of the DNA-based SVM models.

average Recall was 95% and F1-score was 97%. Subsequently, we implemented the trained SVM model on the DNA computing architecture through SDR. We used 50 random samples (35 CMSN samples and 15 normal samples) from the CMSN dataset to verify the ability of the DNA-based SVM model to classify disease states (Fig. 9(c)). Detailed classification outcomes for two representative samples are also provided in Fig. 9(c). These results show that the DNA-based SVM model has good feasibility and can accurately classify CMSN disease states within only two hours, generating uniformly high intensity output signals. We further use the DNA-based SVM model to classify the disease states of 274 samples (240 CMSN samples and 34 normal samples) in the test set (Fig. 9(d)). The test results show that the classification results of the DNA-based SVM model are strong agreement with the actual disease states labeled in TCGA and the classification results of the silicon-based SVM model. Therefore, the accuracy of the DNA-based SVM model on the test set remains consistent with that of the silicon-based SVM, which is 98.5%.

B. Parallel Disease Diagnosis Using the DNA-Based SVM Models

The realization of robust parallel diagnosis of multiple diseases from a single biological sample has attracted much attention in molecular medicine, which can save resources, shorten diagnosis time and improve diagnostic efficiency. To evaluate the parallelism of the DNA-based SVM model, we concurrently diagnosed CMSN, Glioma, and CCC (Fig. 10(a)). In addition to the SVM model mentioned above for CMSN diagnosis, we trained two SVM models for Glioma and CCC. We trained the SVM model on a silicon electronic computer using the miRNAs expression level data of Glioma retrieved from TCGA (531 tumor samples and 95 normal samples), and extracted the disease-relevant characteristic miRNAs (miRNA-219a-2, miRNA-92b, miRNA-9-1, miRNA-891a and miRNA-153-2) together with their associated weight values (0.1, 0.1, 0.17, 0.08 and 0.07), as shown in Fig. 10(b). We calculated six main evaluation metrics of this model: Sensitivity was 99.4%, Specificity was 100%, Precision was 100%, Accuracy was 99.5%, average Recall was 100% and F1-score was 99% (Fig. 10(c)). We trained the SVM model on a silicon electronic computer using the miRNAs expression level data of CCC retrieved from TCGA (538 tumor samples and 73 normal samples), and extracted the disease-relevant characteristic miRNAs (miRNA-584, miRNA-206, miRNA-514b, miRNA-509-3 and miRNA-934) together with their associated weight values (0.83, -0.8, -0.66, -0.85 and -1.33), as shown in Fig. 10(b). We calculated six main evaluation metrics of this model: Sensitivity was 98.8%, Specificity was 91.3%, Precision was 98.8%, Accuracy was 97.8%, average Recall was 95% and F1-score was 95% (Fig. 10(c)). Then, we implemented the three trained SVM models on the DNA computing architecture through SDR. We used the DNA-based SVM model to classify the Glioma disease states of 186 samples (154 Glioma samples and 32 normal samples) in the test set (Fig. 10(d)). We used the DNA-based SVM model to classify the CCC disease states of 183 samples (160 CCC samples and 23 normal samples) in the test set (Fig. 10(e)). The test results show that the classification results of the DNA-based SVM model are strong agreement with the actual disease states labeled in TCGA and the classification results of the silicon-based SVM model. It is worth noting that, in parallel disease diagnosis, the final "healthy" label is not determined by any single model's output; rather, a healthy conclusion is reached if and only if all three models simultaneously emit a healthy signal. Accordingly, a three-input DNA AND gate is designed to serve as the voting module, producing the final health output exclusively upon the coincident reception of healthy votes from each of the three models (Fig. 10(f)). We further used different disease state cases to test the parallelism of the DNA-based SVM models (Fig. 10(g)). The test results show that the three DNA-based SVM models can accurately diagnose various conditions of disease states, including only having disease CMSN, Glioma, or CCC, two of them, three of them, and health. Moreover, even when the diagnostic task scales from one to three diseases, the run-time remains essentially unchanged and matches that of a single-disease diagnosis. This observation further demonstrates the excellent intrinsic parallelism of the DNA-SVM models in DNA computing architecture.

Compared with analog and fuzzy methods for disease diagnosis based on miRNA data under silicon-based models [5], [6], [7], [10], [11], our method offers the following key advantages: 1) Excellent biocompatibility. Silicon-based models rely on manual capture of miRNA expression levels from biological samples, which are then fed back into the models, creating a difficult-to-bridge gap. In contrast, DNA-based methods enable seamless integration between biological samples and classification decision models, eliminating signal distortion and accuracy loss during conversion and realizing intelligent integrated diagnosis; 2) High parallelism. DNA-based methods possess inherent parallelism, enabling facile classification of multiple disease states in a single test tube. In contrast, silicon-based methods necessitate additional parallel computing architectures and hardware support. Nevertheless, silicon-based computing architectures are far more mature than DNA-based computing architectures, and machine learning methods under silicon-based frameworks are more suitable for tasks involving multi-target and large-scale screening, such as miRNA-disease associations (MDAs) and drug screening [8], [9], [12], [13]. Furthermore, compared with several classical methods under DNA computing architectures [22], [23], [24], [25], [41], our method achieves flexible control over weights, easily detectable unified high-intensity output signals, and parallel diagnosis of multiple diseases while ensuring diagnostic accuracy.

V. Conclusion

DNA molecules serve as an excellent interface between biotechnology (BT) and information technology (IT). In this work, we design four functional modules via SDR: weighted summation module, subtraction module, signal restoration module, and reporter module. In the weighted summation module, we use the concentration ratio of two DNA complexes to flexibly control the weight values. We also introduce a signal restoration module between the subtraction module and the reporter module, which can effectively avoid signal attenuation and obtain the uniform high-intensity output signals. Then, we cascade these modules to build the DNA-based SVM models in DNA computing architecture. In a purified biological sample containing all target miRNAs, silicon-based models require manually designed probes to capture target miRNAs and read their expression levels, this process is time-consuming and labor-intensive. In contrast, DNA-based SVM models, leveraging precise base complementary pairing and SDR, can specifically and precisely bind to the target miRNAs, intelligently sense their expression levels in the biological sample, and accomplish disease state classification. Finally, we use the designed DNA-based SVM models to achieve precise parallel diagnosis of CMSN, Glioma, and CCC. Our work provides a new paradigm for engineering precise, intelligent and integrated disease diagnosis schemes at the molecular level. Based on this work, our future research includes: 1) Placing the model training process in DNA computing architecture to achieve the fully integrated of model training and application; 2) Introduce more functional DNA computing modules to achieve the construction of more complex DNA-based machine learning models, such as introducing DNA modules for implementing kernel functions, enabling DNA-based SVM models to efficiently handle nonlinear data; 3) Orthogonal sequence design improved and homologous miRNA interference avoided, the scale and scope of parallel diagnostics are expected to expand; 4) Optimize the design of DNA structure and modules to achieve a more compact DNA-based machine learning models.

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